Feature Report: Geographic Features

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Basic Information

- **Total Customers**: 61,410
- **Total Features**: 28
- **Feature Columns**: country_diversity, province_diversity, city_diversity, cross_border_txn_count, cross_border_txn_pct, cross_border_flag, is_country_financial_hub, is_country_offshore_structure_jurisdiction, is_country_trade_conduit, is_country_private_banking_hub, is_country_shell_company_jurisdiction, is_country_tbml_high_risk, is_country_major_port, is_country_real_estate_ml_risk, is_country_hnwi_concentration, is_country_high_cash_usage, txn_per_unique_location, high_geographic_dispersion, customer_country_financial_hub, customer_country_offshore_structure_jurisdiction, customer_country_trade_conduit, customer_country_private_banking_hub, customer_country_shell_company_jurisdiction, customer_country_tbml_high_risk, customer_country_major_port, customer_country_real_estate_ml_risk, customer_country_hnwi_concentration, customer_country_high_cash_usage

Data Quality Checks

☑ No Null Values

All features have complete data.

☑ No Duplicate Customer IDs

Feature Statistics

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero %
country_diversity	Integer	0	12	2.34	2	1.77	11,979	19.51%
province_diversity	Integer	0	20	6.60	7	4.83	11,980	19.51%
city_diversity	Integer	0	96	16.42	9	17.61	11,972	19.50%
cross_border_txn_count	Integer	0	196	10.83	3	16.36	19,452	31.68%
cross_border_txn_pct	Float	0.00	1.00	0.14	0.14	0.14	19,452	31.68%
cross_border_flag	Integer	0	1	0.68	1	0.47	19,452	31.68%
is_country_financial_hub	Integer	0	1	0.39	0	0.49	37,707	61.40%
is_country_offshore_structure_jurisdiction	Integer	0	1	0.13	0	0.34	53,429	87.00%
is_country_trade_conduit	Integer	0	1	0.08	0	0.28	56,219	91.55%
is_country_private_banking_hub	Integer	0	1	0.07	0	0.26	56,958	92.75%
is_country_shell_company_jurisdiction	Integer	0	1	0.19	0	0.39	49,921	81.29%
is_country_tbml_high_risk	Integer	0	1	0.08	0	0.28	56,219	91.55%
is_country_major_port	Integer	0	1	0.08	0	0.28	56,219	91.55%
is_country_real_estate_ml_risk	Integer	0	1	0.80	1	0.40	12,360	20.13%
is_country_hnwi_concentration	Integer	0	1	0.07	0	0.26	56,958	92.75%
is_country_high_cash_usage	Integer	0	1	0.04	0	0.20	58,802	95.75%
txn_per_unique_location	Float	0.00	15.00	1.64	1.53	1.16	11,972	19.50%
high_geographic_dispersion	Integer	0	1	0.67	1	0.47	20,221	32.93%

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero %
customer_country_financial_hub	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_offshore_structure_jurisdiction	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_trade_conduit	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_private_banking_hub	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_shell_company_jurisdiction	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_tbml_high_risk	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_major_port	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_real_estate_ml_risk	Integer	1	1	1.00	1	0.00	0	0.00%
customer_country_hnwi_concentration	Integer	0	0	0.00	0	0.00	61,410	100.00%
customer_country_high_cash_usage	Integer	0	0	0.00	0	0.00	61,410	100.00%

Validation Checks

☑ No Unexpected Negative Values

☑ No Infinite Values

Outlier Detection (Values > 3 Standard Deviations)

- country_diversity: 314 outliers (0.51%)
- city_diversity: 388 outliers (0.63%)
- cross_border_txn_count: 1,271 outliers (2.07%)
- cross_border_txn_pct: 719 outliers (1.17%)
- is_country_trade_conduit: 5,191 outliers (8.45%)
- is_country_private_banking_hub: 4,452 outliers (7.25%)
- is_country_tbml_high_risk: 5,191 outliers (8.45%)
- is_country_major_port: 5,191 outliers (8.45%)
- is_country_hnwi_concentration: 4,452 outliers (7.25%)
- is_country_high_cash_usage: 2,608 outliers (4.25%)

... and 1 more features with outliers

Flag Feature Distributions

Feature	True Count	True Percentage	False Count	False Percentage
cross_border_flag	41,958	68.32%	19,452	31.68%

Summary

- ☑ **Total Features**: 28
- ☑ **Data Completeness**: 3.57%
- ☑ **Customers Covered**: 61,410
- ☑ **Flag Features**: 1

Report generated automatically by feature engineering pipeline